

StoreFlow AI: Transforming Retail Commerce

Physical stores drive \$25T globally but operate blind - StoreFlow AI is "Waze for inside the store"

CONTINUOUS OPERATING LOOP (OBSERVE -> DIAGNOSE -> PRESCRIBE -> VERIFY)



THE LEGACY IMPASSE

- **Extreme Costs:** High sensor costs (\$1,200/sensor, \$10,000+/store CAPEX).
- **Blind Operations:** Legacy retail operates completely blind vs e-commerce analytics.
- **Privacy Penalties:** Strict liability under India DPDP Act 2023 & EU GDPR Article 9.

STOREFLOW AI BREAKTHROUGH

- **Edge Hardware:** Extracts spatial intelligence via a low-cost \$130 edge appliance.
- **Zero CAPEX:** Reuses existing legacy CCTV infrastructure directly.
- **100% Zero-PII Engine:** Stream processing with zero privacy risk or compliance friction.

\$0

New Camera CAPEX

14,000x

Bandwidth Reduction

>97%

TCO Reduction

COMPUTER VISION DATA PIPELINE ARCHITECTURE

TIER 1: INGESTION

PHYSICAL INGESTION

8-16 RTSP Feeds

Hardware Decoding
NV12 Frames in Unified DMA
Memory

Zero-Copy Ingest:

Maintains NV12 frames
directly in DMA memory to
eliminate CPU overhead.

LOCAL NVMe SSD

Offline Resilience
Spools up to 90 days
operational data

TIER 2: EDGE PROCESSING

STOREFLOW EDGE APPLIANCE (\$130 Mini-PC)

- YOLOv10-Nano INT8 + OC-SORT
> Latency: 1.84ms
- Planar Homography
- OSNet-0.5x Embeddings
(Face-masked for disjoint handoffs)
- Walking-Velocity Graph Priors

MQTT / TLS TELEMETRY

Anonymous 65-byte trajectory JSON tokens
Bandwidth: < 3.5 KB/s

TIER 3: ANALYTICS

SAAS ANALYTICS HUB

ClickHouse / DuckDB

- Gaussian KDE
- Savitzky-Golay Dwell Filtering
- Markov Chain Dead-Zone
Discovery

1.84ms

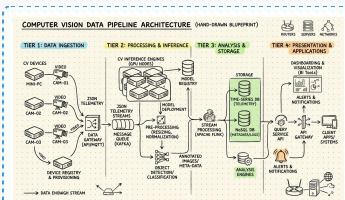
INFERENCE LATENCY

TIER 4: PRESENTATION

PRESENTATION LAYER

Interactive WebGL Cockpit

Light-Mode UI with real-time
trajectory visualization & spatial
heatmaps.



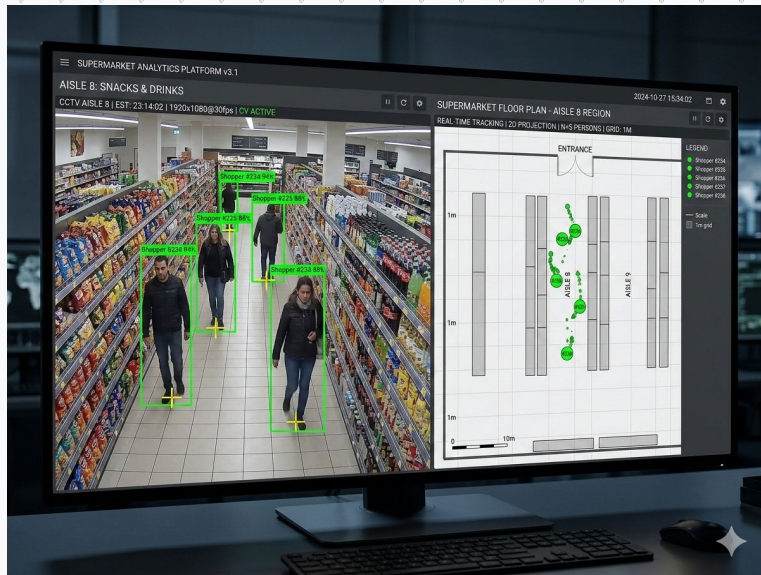
[REFERENCED ARCHITECTURE
BLUEPRINT SKETCH]

Spatio-Temporal Localization & Re-ID

Parallax-Free Tracking, Zero-PII Embeddings & Geodesic Link Modeling

STOREFLOW AI //
ARCHITECTURE

LIVE CCTV DETECTION PROOF // AISLE 8 REGION



> 100% ZERO-PII COMPLIANCE PROTOCOL

Upper 25% of bounding boxes masked to zero prior to OSNet-0.5x 128-d embedding extraction, completely eliminating facial biometrics.

01 / PARALLAX-FREE PLANAR HOMOGRAPHY (H)

- **Ankle-Keypoint Regression:** Localizes ground contact points directly.
- **Zero-Touch VP Calibration:** Calculates focal length & tilt angle (θ) via absolute conic orthogonality constraints.

<0.11m

METRIC ACCURACY FOR
GROUND CONTACT

02 / SPATIO-TEMPORAL CAMERA LINK MODEL (CLM)

- **Geodesic Velocity Bounds:** Models human walking physics (0.8 - 2.5 m/s) along store aisle paths.
- **Blind Spot Tracking:** Links re-identification across disjoint camera fields of view seamlessly.

86.4%

IDF1 TRACKING SCORE

Key Takeaway: Combining ankle keypoint homography with human kinematics guarantees continuous shopper journeys across complex supermarket layouts without relying on biometric identifiers.

Precision Dwell & Store Friction Analytics

Spatial engagement, bottleneck detection, and dead-zone discovery

LIVE FLOORPLAN ANALYTICS // VISUAL PROOF



Produce Micro-Zones

1.0m Voronoi grid isolating engagement from transit

Checkout Choke Hotspot

Spillover alert triggered in <35 seconds

1. High-Precision Shelf Dwell Engine

Separates trajectory transit from true product engagement utilizing **Savitzky-Golay trajectory smoothing** and **1.0m Voronoi cells**.

- > Computes Engagement-to-Passby Ratio
- > Family & group filtering for true density

2. Fluid Dynamics & Store Friction Score

Applies fluid dynamics equations to model traffic choke points and generate a **Store Friction Score (0-100 RAG Index)**.

- > Alert: Queue spillover predicted within 35s

3. Markov Chain Dead-Zone Discovery

Calculates **Spatial Opportunity Scores** to automatically identify flow drop-offs across cold zones.

- > Auto-highlights dead aisles

<8%

Discovery
Rate Flag

2D Digital Twin & Spatial Intelligence Cockpit

Automated CAD vectorization, continuous heatmaps & physical A/B testing engine



01 // 2D Digital Twin & Blueprint CAD

Automatically converts CAD DXF, SVG, or PDF blueprints into an interactive digital twin with semantic fixture zoning.

02 // Gaussian Heatmaps & Traces

Continuous density heatmaps reveal browsing zones & checkout choke points with directional customer navigation traces.

03 // 30-Sec Mobile Dispatch

Managers receive real-time, 1-tap quick action alerts for store bottlenecks or dead-zones via WhatsApp, Telegram, or App Push.

> ENGINE STATUS: CAD AUTO-VECTORIZATION ACTIVE

- DXF / SVG / PDF blueprints automatically mapped to semantic fixture zones.
- Executive cockpit streams real-time foot-traffic trajectories & dwell heatmaps.

04 // Physical Store A/B Verification

Layout experimentation engine provides concrete before-and-after verification reports on traffic lift, friction reduction, & sales conversion.

StoreFlow AI: Zero-Disruption Deployment

Operational intelligence live in 4 seamless, non-intrusive steps

1

30m

Blueprint Ingestion

Ingest and parse existing blueprints or fire-evacuation maps.

Converts architectural layouts into semantic vector representations automatically.

2

15m

CCTV Auto-Discovery

ONVIF WS-Discovery registers camera frustums in under 45 seconds.

Full camera discovery and spatial alignment completed in 15 minutes.

3

10m

Plug-In Edge Box

Connect edge appliance directly into an existing PoE switch.

> Requires 12W power
> 0 firewall ports open

4

48h

Go-Live in 48h

48-hour baseline learning establishes walking velocity and queue norms.

Manager cockpit triggers active live analytics post-baseline.



✓ Guaranteed Zero Store Downtime & No Network Configuration Overhaul

Enterprise Viability & Statutory Compliance

Designed for Thin-Margin Retail // Hybrid B2B SaaS Model

TCO COMPARISON: LEGACY VS STOREFLOW AI

COMPONENT	LEGACY SYSTEM	STOREFLOW AI
Cameras	Specialized IP / 4K	Existing CCTV
Edge Hardware	Heavy Server (\$8k+)	Rs 15k-35k (\$180-\$420)
Bandwidth Efficiency	High Video Stream	3.25 KB/s Extreme
Software / Store	\$1,500 - \$2,500/mo	Rs 4,999 (\$60)/mo
Cloud GPU OPEX	Heavy Cloud Streaming	Zero Cloud GPU
Year 1 TCO / Store	\$24,000 - \$35,000	\$601 - \$969

>97% TCO

COST REDUCTION

< 3 WEEKS

FULL PAYBACK PERIOD

PRIVACY-BY-DESIGN & COMPLIANCE



100% Zero-PII Guaranteed

Video frames processed entirely in volatile GPU DMA memory .

- Zero facial biometric storage
- Zero raw video persisted
- Instant frame purge post-inference

INDIA STATUTORY

DPDP Act 2023

Fully Compliant Architecture

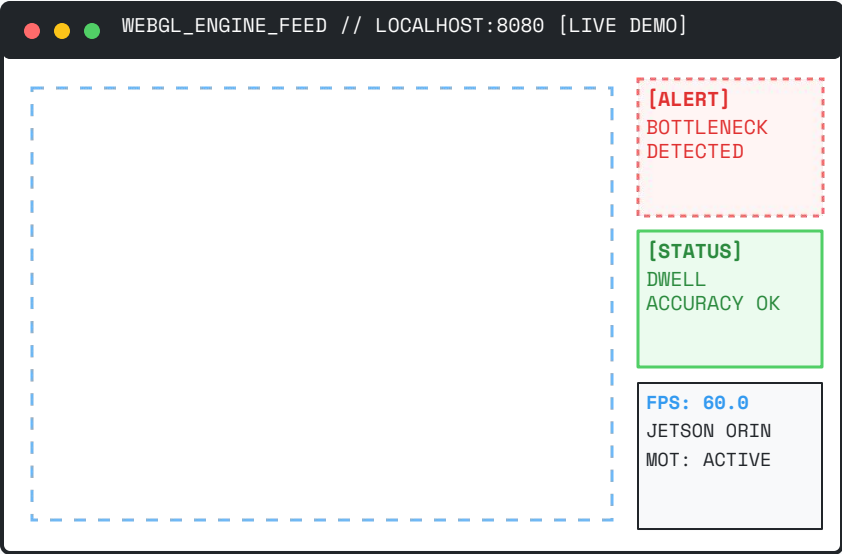
EU STATUTORY

GDPR Article 9

Special Category Exempt

Bandwidth Breakthrough: Operating at just 3.25 KB/s per store allows deployment over 4G/5G dongles without expensive leased lines.

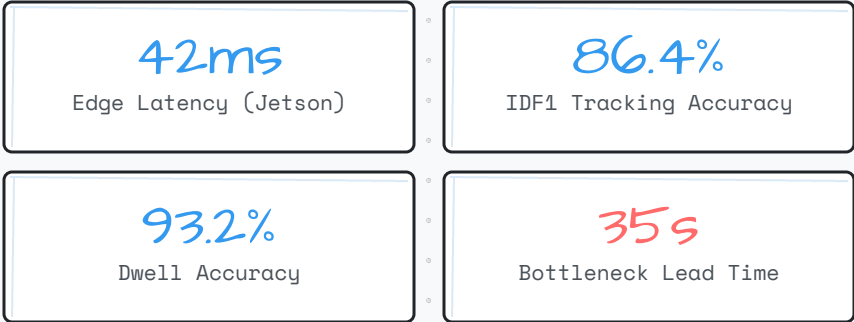
Empirical Validation & Commercial Roadmap



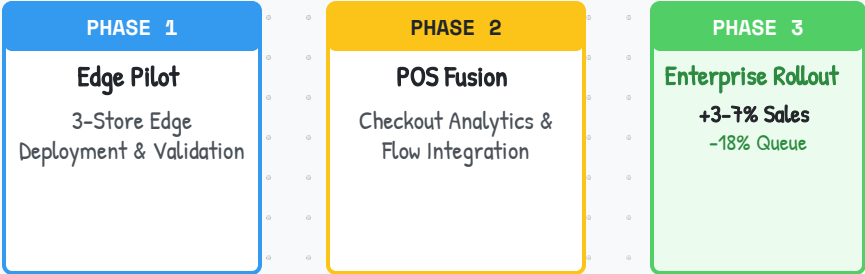
100% Local, Self-Contained Edge Demo Stack

Runs entirely on localhost, ensuring a zero-internet fail-safe for live pitch demonstrations. Complete isolation from cloud dependency guarantees rock-solid reliability during live stakeholder trials.

Empirical Benchmark Scorecard



90-Day Commercial Rollout Roadmap



TARGET IMPACT: +3-7% Top-Line Sales Lift | 18% Queue Time Reduction